

Limit Question

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Ques 1: Find the value of the infinite sum $\sum_{k=1}^{\infty} \frac{1}{k} \left(\frac{1}{n^{k+1}} \sum_{m=1}^m m^k \right)$
Consider the partial sum of that series; let us assume that it is

$$S_k = a_1 + a_2 + \dots + a_k$$

where $a_k = \frac{1}{k} \lim_{n \rightarrow \infty} \frac{1}{n^{k+1}} \sum_{m=1}^n m^k$ so, if we can find the value of S_k for every k , then we can find the value of sum, first we evaluate the limit part of S_k by using limit of sum concept
consider, $l = \lim_{n \rightarrow \infty} \frac{1}{n^{k+1}} \sum_{m=1}^n m^k$

$$= \sum_{m=1}^{\infty} \left(\frac{m}{n} \right)^k * \frac{1}{n}$$

$$= \int_0^1 x^k dx$$

after solving this, we get

$$= \left[\frac{x^{k+1}}{k+1} \right]_0^1$$

$$= \left(\frac{1^{k+1}}{k+1} \right) - \left(\frac{0^{k+1}}{k+1} \right)$$

$$= \frac{1}{k+1}$$

hence, $a_k = \left(\frac{1}{k} \right) \cdot \left(\frac{1}{k+1} \right)$

Now, $S = \lim_{k \rightarrow \infty} S_k = \lim_{k \rightarrow \infty} \sum_{i=1}^k \frac{1}{i \cdot (i+1)}$

$\sum_{i=1}^k \frac{1}{i \cdot (i+1)} = \sum_{i=1}^k \left(\frac{1}{i} - \frac{1}{i+1} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{k-1} - \frac{1}{k} \right) + \left(\frac{1}{k} - \frac{1}{k+1} \right)$
so, after cancel the terms we get

$$= 1 - \frac{1}{k+1}$$

hence, $S = \lim_{k \rightarrow \infty} S_k = \lim_{k \rightarrow \infty} \left(1 - \frac{1}{k+1} \right) = 1$

$$\boxed{S = 1}$$